TÉCNICO
UNIVERSIDADE
DE LISBOA

0.2 · SHIP AND OCEAN SYSTEMS DESIGN

# Ship Design Project

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# Structure of the Project

Ten **common deliverables (D1–D10)**, extended by **two (2) design specializations** per group.

The project is group work done outside class. The in-class exercises are individual practice on a reference ship and are not graded.

COMMON TO THE WHOLE PROJECT

**Baseline Project (A or B)**

E.G. GROUP 1 · BASELINE A

**Preliminary Stability + Damage Probabilistic Calculations**

E.G. GROUP 2 · BASELINE B

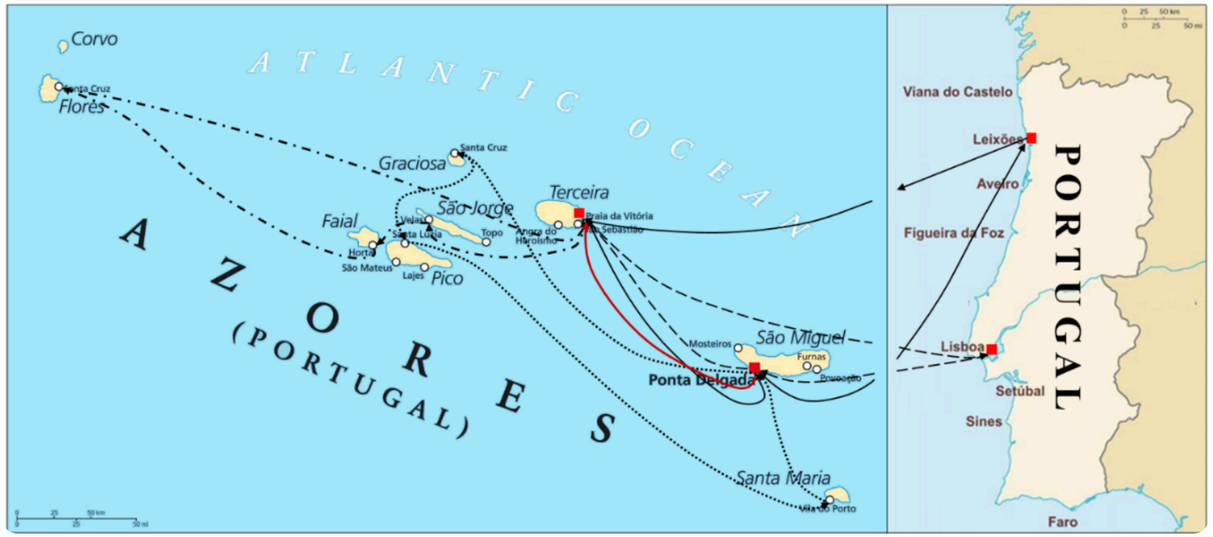
**Electrical Load Balance + Alternative Fuel and Energy Storage**

## AVAILABLE DESIGN SPECIALIZATIONS

| | | | |
|---|---|---|---|
| **SP01** Longitudinal Strength Calculation | **SP02** Hull–Propeller–Engine Matching | **SP03** Electrical Load Balance | **SP04** Detailed Cost Estimation |
| **SP05** Parametric Hull Shape Generation | **SP06** Engine Room and Steering Room Arrangement | **SP07** Preliminary Stability Calculation | **SP08** Hull Shape Optimization |
| **SP09** Damage Probabilistic Calculations (Baseline A) | **SP10** Detailed Emissions Calculation | **SP11** Advanced Dimensional Analysis | **SP12** Alternative Fuel and Energy Storage |

# Two Ships for One Azores Container Service

Each group designs one of the two ships, assigned by the lecturer.







## Baseline A · Azores Express

1 300 TEU · mainland to islands, weekly · dual-fuel LNG




## Baseline B · Azores Feeder

130 TEU · inter-island loops · battery-electric

# Azores Express: Owner’s Requirements

### Ship type
Container ship, forward superstructure, aft engine room, bow thruster, good seakeeping

### Cargo
Containers · gearless

### Target capacity
1 300 TEU

### Route
Lisbon – Ponta Delgada – Praia da Vitória – Lisbon, one round voyage a week

### Port restrictions
Probably none · to be checked by each group

### Service speed
18–19 kn · 7–8 kn on the generators through a PTI

### Fuel and propulsion
Dual-fuel HFO/LNG, single shaft · LNG in tanks or tank containers aft

### Rules and regulations
Bureau Veritas · Portuguese flag · IMO conventions and codes

### Design objective
Minimum propulsion power, with one round voyage per week

### Environmental targets
EEDI · CII · ECA on the route

### Specialization 1
Per group

### Specialization 2
Per group

**Need:** replaces an ageing fleet on the line. **Alternative route:** Leixões – Praia da Vitória – Ponta Delgada – Leixões, also one round voyage per week.

# Azores Feeder: Owner’s Requirements

### Ship type
Container ship on a large PSV-type hull (X-bow?), forward superstructure, bow thruster, good seakeeping

### Cargo
Containers · geared, with its own container crane

### Target capacity
130 TEU

### Route
Two island loops plus one transit, all in one week (below)

### Port restrictions
Probably none · to be checked by each group

### Service speed
11–12 kn · 6–7 kn on the generator sets, to the mainland or in emergency

### Fuel and propulsion
Swappable battery containers, aft on deck · two electric motors and shaft lines

### Rules and regulations
Bureau Veritas · Portuguese flag · IMO conventions and codes

### Design objective
Minimum battery capacity, with both loops in one week

### Environmental targets
EEDI · CII · ECA only on the mainland transit

### Specialization 1
Per group

### Specialization 2
Per group

**Need:** replaces an ageing fleet on the island lines. **Loop 1:** Ponta Delgada – Graciosa – Pico – Santa Maria – Ponta Delgada · **Loop 2:** Praia da Vitória – São Jorge – Faial – Flores – Praia da Vitória · plus the transit Ponta Delgada – Praia da Vitória. Batteries sized for one loop plus the transit.

# Common Deliverables D1–D10

Deliver the work in a single ZIP file (Ex.: Group1_Project.zip), by e-mail, until 13 December 2026.

| # | CONTENT | MUST DEMONSTRATE |
|---|---|---|
| D1 | **Design Basis** | Owner's requirements · Port limits · Environmental targets |
| D2 | **Reference Ship Database** | Recent similar ships · Sources · Regressions and bounds |
| D3 | **Ship Model** | Inputs and outputs · Calculation sequence · Closed weight balance |
| D4 | **Voyage and Financial Model** | Ship model outputs as input · Port restrictions · Revenue assumptions |
| D5 | **Validation Note** | Reference ships · Error per quantity · Model corrected |
| D6 | **Optimization Report** | Justified objective · Active constraints · Optimum vs requirements |
| D7 | **Compliance Sheet** | EEDI and CII · Freeboard and stability · Draught at every port |
| D8 | **General Arrangement and Hull Form** | A3 sketch: profile and midship section · Hydrostatics check |
| D9 | **Specialization Documents (2)** | Calculation note · Required drawings · One per specialization |
| D10 | **Datasheet and Presentation** | One-page A4 datasheet · Presentation of 15 minutes |

**Note:** List of detailed deliverables will be handed out to each group on Fri 13 Nov.

**Note:** The class exercises are practice, not part of the project.

# Specialization Content

| # | DESIGN SPECIALIZATION | MUST DEMONSTRATE |
|---|---|---|
| SP01 | **Longitudinal Strength Calculation** | Loads and bending moments · Section modulus check · Midship section drawing |
| SP02 | **Hull–Propeller–Engine Matching** | Optimum B-series propeller · Engine matching · PTO/PTI modes · Stern drawing |
| SP03 | **Electrical Load Balance** | Consumers per operating mode · Generator sizing · One-line diagram |
| SP04 | **Detailed Cost Estimation** | Cost breakdown by group · Unit cost sources · Link to General Arrangement |
| SP05 | **Parametric Hull Shape Generation** | Parametric hull definition · Coupling to the optimizer · Lines plan |
| SP06 | **Engine Room and Steering Room Arrangement** | Machinery layout · Rudder and steering gear · Arrangement drawing |
| SP07 | **Preliminary Stability Calculation** | Loading conditions · GZ curves against criteria · Tank and capacity plan |
| SP08 | **Hull Shape Optimization** | Hull variables and objective · Resistance gain · Initial and optimized lines |
| SP09 | **Damage Probabilistic Calculations** | Baseline A only · Watertight subdivision · A ≥ R · Subdivision plan |
| SP10 | **Detailed Emissions Calculation** | Fuel use per mode · CO2, SOx, NOx per year · Machinery arrangement |
| SP11 | **Advanced Dimensional Analysis** | Regression equations for dimensions and ratios · Fit quality and validity range |
| SP12 | **Alternative Fuel and Energy Storage** | Energy per voyage · Tank or battery sizing · Rules check |

**Note:** List of detailed deliverables will be handed out to each group on Fri 13 Nov.

**Note:** Each specialization has its own self-study deck.

**Note:** Your group's two are written on its Project Scope Sheet.

RESOURCES

# Software

## Calculations and programming

**Python**

High-level programming language

**Google Colab or VS Code**

Integrated development environment (IDE)

**Excel**

Spreadsheet and Solver, if Python is not used

## 3D modelling and drafting

**Rhinoceros 3D**

Surface modeling

**Maxsurf Modeler**

Hull form modeling

**DELFTship (free)**

Hull form modeling

**AutoCAD**

Computer Aided Design system

**Note:** Bring a computer to every class: each student does the exercises individually, in Python or Excel.

# Groups and Questions

## 1

### Form your group

Start now. Groups are final on **Fri 6 Nov**.

## 2

### Ship and specializations

The lecturer assigns each group a Baseline ship (A or B) and two specializations by **13 Nov**. Keen on a particular ship type? Make your case.

## 3

### Your questions

On the project, now. Later: office hours, one 30-minute slot per group per week, at calendly.com/vnsd-luisbatista/meetme.

Everything for the course is on the course site: l-u-i-s-b-a-t-i-s-t-a.github.io/PNSO-2026

**Note:** 0.3 Report Template, 0.4 Datasheet Template and 0.5 Project Scope Sheet are provided on Fri 13 Nov.